

Sales Trend and Time-Based Performance Analysis for Afficionado Coffee Roasters

1. Introduction

Afficionado Coffee Roasters aims to understand customer purchasing behavior based on time and store location. This project analyzes transaction-level sales data to identify peak sales periods, busiest hours, and store performance patterns. The insights help management make data-driven decisions regarding staffing, inventory management, and operational planning.

2. Problem Statement

The company lacks a consolidated view of sales trends and customer demand patterns. Without proper analytics, staffing and operational decisions are based on assumptions rather than actual data. This project addresses these challenges through time-based sales analysis.

3. Dataset Description

The dataset contains transaction-level coffee sales records with the following fields:

- transaction_id
- year
- transaction_time
- transaction_qty
- unit_price
- store_id
- store_location
- product_id
- product_category
- product_type
- product_detail

Additional features created during analysis:

- Revenue
- Hour
- Day of Week

4. Methodology

1. Loaded the dataset using Pandas.
2. Calculated revenue using:
$$\text{Revenue} = \text{transaction_qty} \times \text{unit_price}$$
3. Converted transaction time into datetime format.
4. Extracted hourly information.
5. Performed hourly revenue analysis.

- 6. Performed store-wise revenue analysis.
- 7. Generated heatmaps for location-based demand comparison.
- 8. Built an interactive Streamlit dashboard.

5. Analysis and Results

5.1 Dashboard Overview

Afficionado Coffee Roasters

Sales Trend and Time-Based Performance Analysis

Total Revenue

\$698,812.33

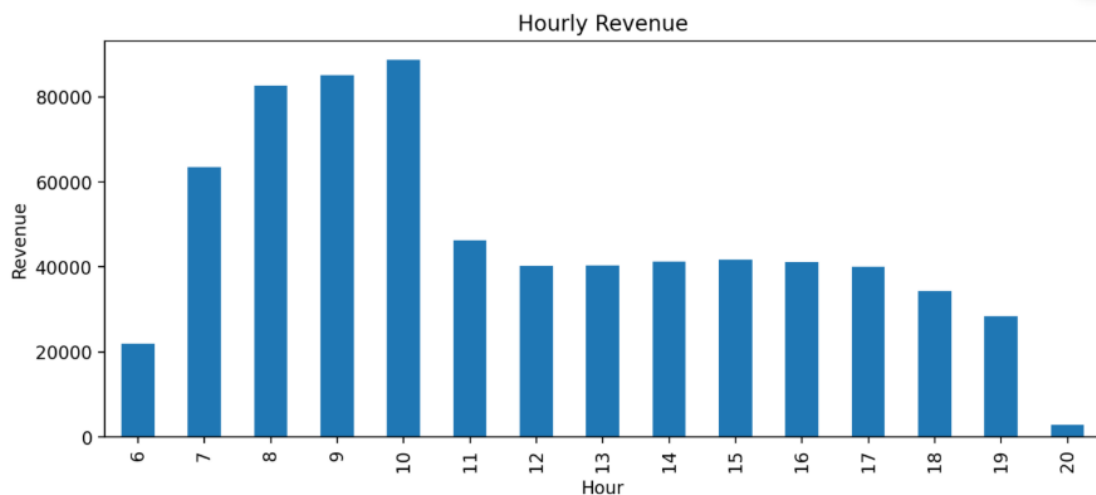
Peak Revenue Hour

10:00

Top Store

Hell's Kitchen

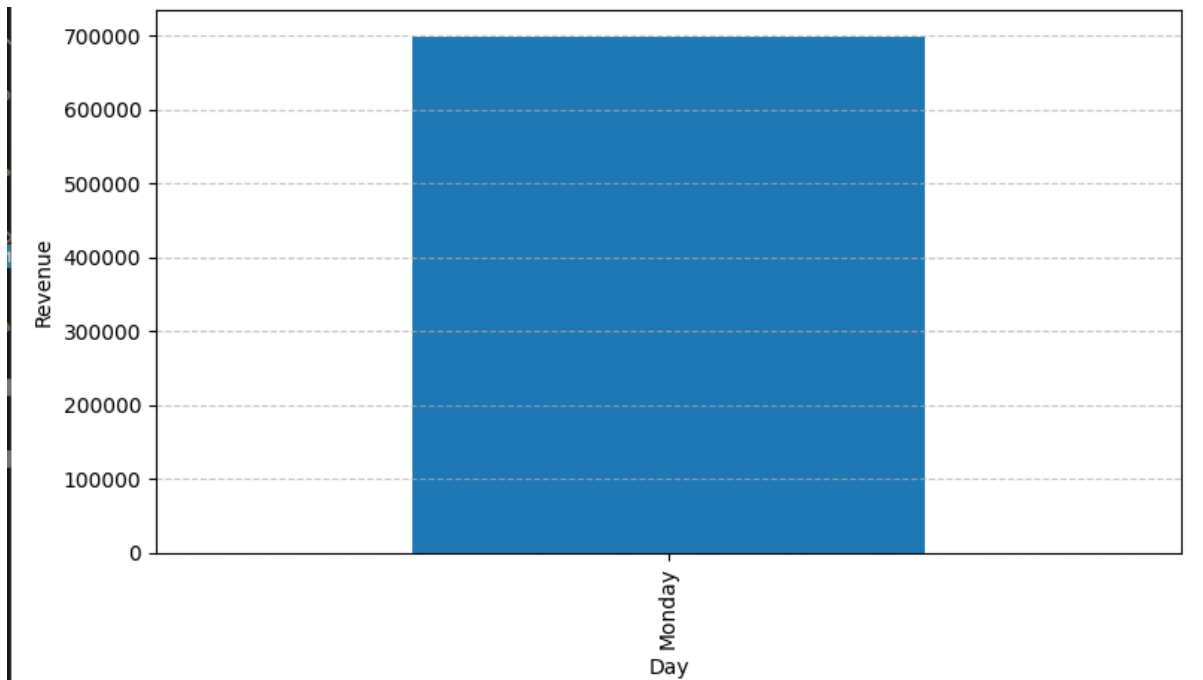
5.2 Hourly Revenue Analysis



The highest revenue was observed at 10:00 AM.

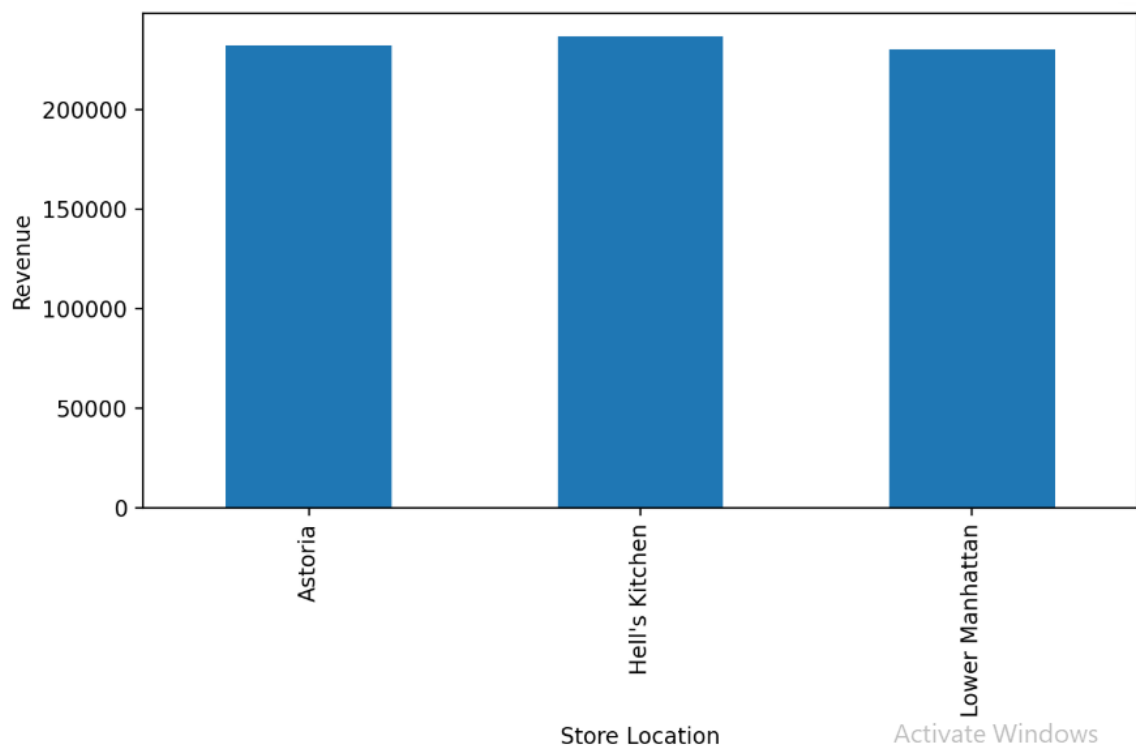
Revenue Generated:
88,673.39

5.3 Revenue by Day of Week



The day-wise revenue chart was generated from the available data. However, since the dataset does not contain actual transaction dates, the day-of-week analysis is limited and should be interpreted with caution.

5.4 Store Performance Analysis



Store-wise revenue analysis was performed to compare the performance of different store locations.

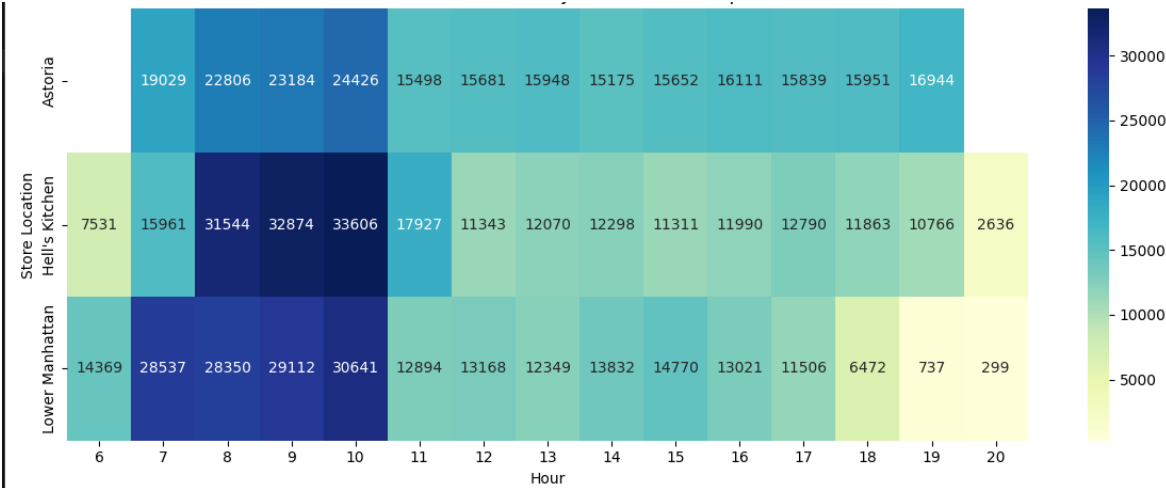
Revenue Ranking:

Store Location	Revenue
Hell's Kitchen	236,511.17
Astoria	232,243.91
Lower Manhattan	230,057.25

Observation:

Hell's Kitchen generated the highest revenue among all store locations and was identified as the top-performing store. Astoria and Lower Manhattan also performed well, with only a small difference in total revenue.

5.5 Store Location vs Hourly Revenue Heatmap



The heatmap highlights revenue patterns across store locations and hours, showing stronger sales during morning periods.

6. Business Insights

- Customer demand is highest between 8:00 AM and 10:00 AM.
- 10:00 AM generates both the highest revenue and transaction volume.
- Sales decline gradually after 11:00 AM.
- Hell's Kitchen is the highest-performing store location.
- Morning periods represent the primary sales window.

7. Recommendations

- Schedule additional staff during peak morning hours.

- Maintain sufficient inventory before the morning rush.
- Use successful strategies from Hell's Kitchen across other stores.
- Introduce promotions during off-peak hours to increase customer traffic.
- Continue using data-driven planning for staffing and operations.

8. Conclusion

The analysis successfully identified important temporal sales patterns across Afficionado Coffee Roasters locations. The results indicate that morning hours, particularly around 10:00 AM, contribute the highest sales and transaction volumes. These insights can support better staffing decisions, improved customer service, and more efficient business operations.